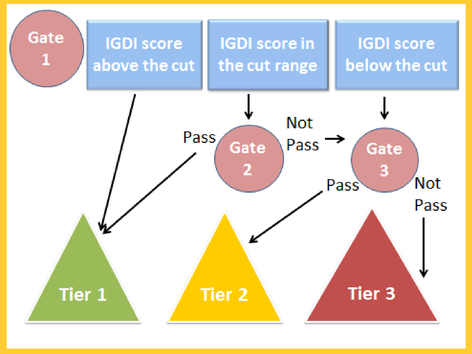
Confident Decision Making: Validation of an Early Childhood Response to Intervention Decision Making Framework

Tracy Bradfield, Kory Vue, Michael Rodriguez, & Scott McConnell

Background

- Use of a Response-to-Intervention (RTI) model requires early educators to identify students requiring more intensive levels of instruction. To do this, they require an efficient and valid assessment system.

- We have developed a Decision-Making Framework (DMF) to identify children requiring tier two and tier three intervention to support their early language and literacy development.



- The DMF is a multiple-gating model including three successive levels or "gates" of decision making.

- The first gate consists of universal screening with the Identification IGDIs 2.0, universal screening scales constructed specifically to maximize information about individual children's performance at a seasonal cut score.

- The ID IGDIs 2.0 measure performance in the domains of oral language development, phonological awareness, alphabet knowledge, and comprehension.

- The 2nd and 3rd gates use results from a researcher-developed Teacher Questionnaire to provide additional information to inform instructional tier placement.

Methods and Results

The study was designed to examine the degree to which the DMF can provide a more efficient yet accurate method of identifying children who require higher levels of instruction compared to the use of an extensive standardized criterion measure.

Research Questions

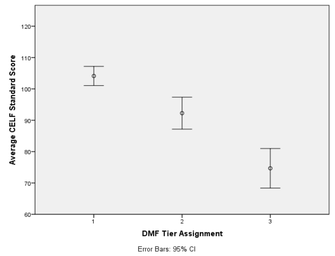
a) What is the relation between scores on the teacher questionnaire and scores on the standardized criterion measure?
b) What is the classification accuracy of the DMF when the standardized measure is used as the criterion of need?
c) Does the mean standardized criterion test score differ significantly across tier assignment groups (tier 1, tier 2, tier 3)?
d) Which variables or combination of variables capture the most variance in predicting language and literacy status?

Oral Language

CELF Assigned Tier vs OL DMF Assigned Tier

		OL DMF Tier Assignment			Total
		1	2	3	
CELF Tier Assignment	1	48	25	4	77
	2	2	8	9	19
	3	1	5	12	18
Total		51	38	25	114

Note: Association is significant, p<.01; Kappa = .58.



Difference in Average CELF Preschool 2 Score across DMF Tier Assignment Groups

DMF Tier Assignment	DMF Tier Assignment	Mean Difference	SE	Cohen's D
1	2	11.3*	2.9	0.6
	3	29.4*	3.4	1.7
2	3	18.2*	3.5	1.0

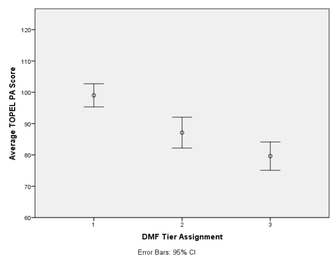
* p<.01

Phonological Awareness

TOPEL PA Assigned Tier vs PA DMF Assigned Tier

		TOPEL PA Tier Assignment			Total
		1	2	3	
PA DMF Tier Assignment	1	52	8	2	62
	2	12	11	5	28
	3	8	12	13	33
Total		72	31	20	123

Note: Association is significant, p<.01; Kappa = .53.



Difference in Average TOPEL PA Score across DMF Tier Assignment Groups

DMF Tier Assignment	DMF Tier Assignment	Mean Difference	SE	Cohen's D
1	2	11.9*	3.1	0.7
	3	19.4*	2.9	1.2
2	3	7.5	3.5	0.5

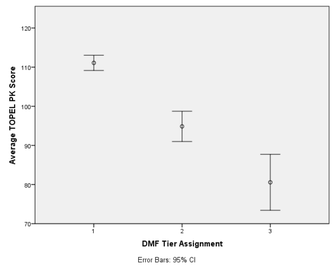
* p<.01

Alphabet Knowledge

TOPEL PA Assigned Tier vs PA DMF Assigned Tier

		PK DMF Tiers			Total
		1	2	3	
TOPEL PK Tiers	1	65	32	1	98
	2	1	17	4	22
	3	0	1	2	3
Total		66	50	7	123

Note: Association is significant, p<.01; Kappa = .49.



Difference in Average TOPEL PK Score across DMF Tier Assignment Groups

DMF Tier Assignment	DMF Tier Assignment	Mean Difference	Std. Error	Cohen's D
1	2	16.2*	2.0	1.1
	3	30.5*	4.2	2.1
2	3	14.3*	4.3	1.0

* p<.01

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Results

Oral Language

- Relation between OL TQ & CELF, r = .64.
- Regression*:
  - Gate 1: IGDI – CELF, r^2 =.40.
  - Gate 2: OL TQ – CELF, r^2 =.11.
  - Gate 3: OL TQ – CELF, r^2 =.29.

Phonological Awareness

- Relation between PA TQ & TOPEL PA, r = .53.
- Regression*:
  - Gate 1: IGDI – TOPEL PA, r^2 =.30.
  - Gate 2: OL TQ – TOPEL PA, r^2 =.11.
  - Gate 3: OL TQ – TOPEL PA, r^2 =.8.

Alphabet Knowledge

- Relation between AK TQ & TOPEL PK, r = .71.
- Regression*:
  - Gate 1: IGDI – TOPEL PK, r^2 =.29.
  - Gate 2: OL TQ – TOPEL PK, r^2 =.49.
  - Gate 3: OL TQ – TOPEL PK, r^2 =.25.

* All regressions were statistically significant.

Discussion

The results here demonstrate that, for each domain, the DMF provides a valid and efficient system for making instructional decisions about children requiring additional levels of language and literacy instruction within an RTI framework.

The Teacher Questionnaire accounts for a significant amount of variance in criterion test score for each domain.

For each domain, the DMF sorts children into tiers which demonstrate significant difference on average criterion test score, with moderate to large effect sizes.

How to define "true need" for tier of instruction was a limitation in this study and a question to be defined by the field.